



Prevalence and Characteristics of Asymmetric Hearing Loss in Korean Adults: A KNHANES 2022–2023 Analysis

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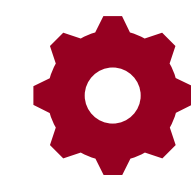
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INTRODUCTION

- Asymmetric Hearing Loss (AHL) is defined by interaural threshold difference.
- National epidemiologic data on AHL in Korean adults remain limited.
- This study evaluated prevalence and audiometric characteristics of AHL using **KNHANES 2022–2023**.
- This study aims to estimate the **prevalence of AHL** in Korean adults aged ≥ 40 years and to **characterize** its age, frequency, laterality, and degree patterns.



MATERIALS & METHODS

- Definition:** Absolute interaural difference ≥ 15 dB HL in 4-frequency average (0.5, 1, 2, 4 kHz), based on the AAO–HNS criterion
- Adults aged ≥ 40 years
- Final sample: N = 3299
 - Weighted population: ~ 20.2 million
- AHL case subset: n = 207



RESULTS ► KEY FINDINGS



6.3%

Weighted prevalence of AHL



4 kHz

Most common asymmetry frequency



L > R

Left-sided AHL more common than right



15–20 dB HL

Largest category of interaural difference

Weighted Prevalence of AHL by Age Group (N = 3299)

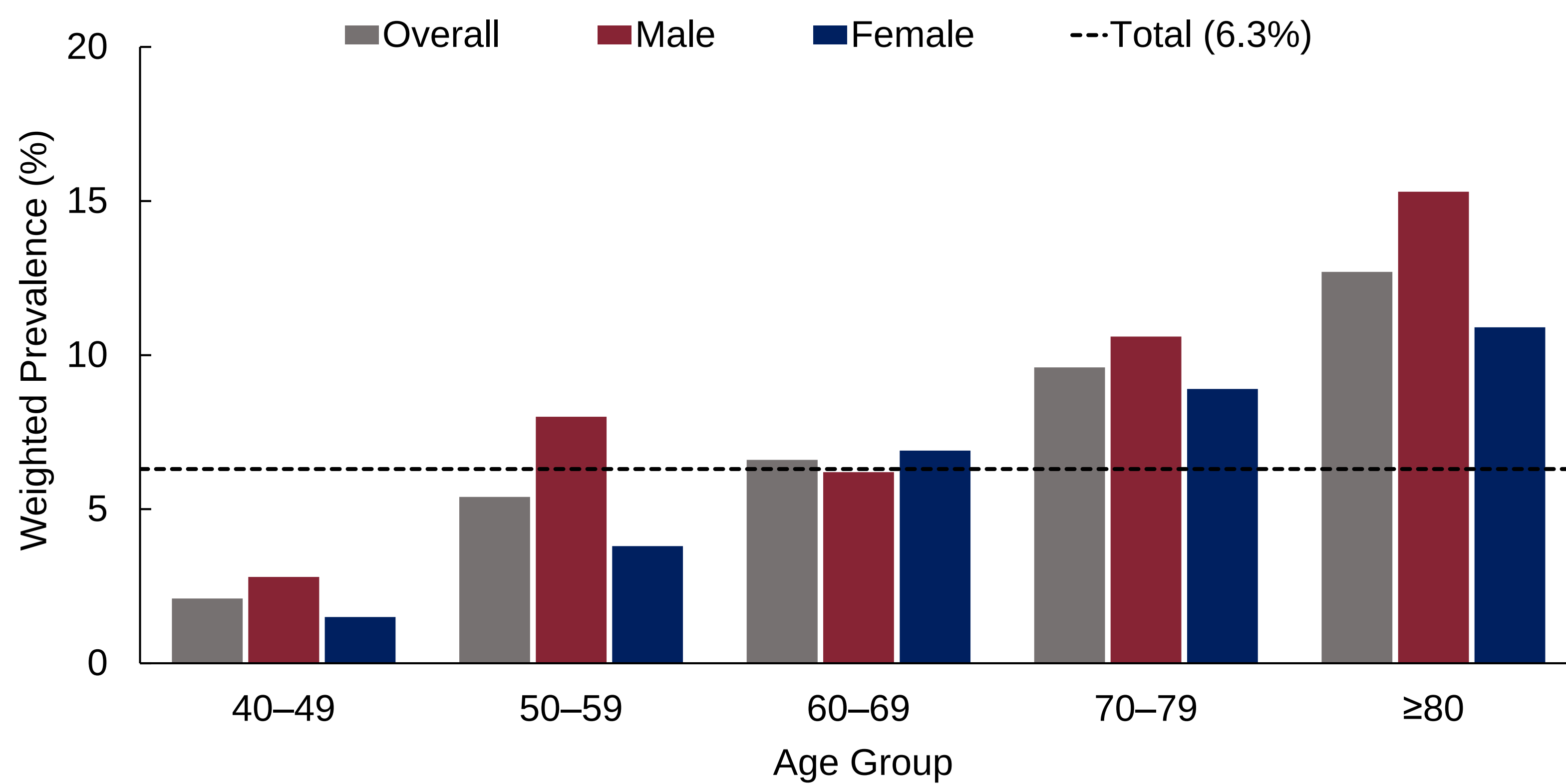


Figure 1. Weighted prevalence of AHL increased with age overall and in both sexes. Dashed line indicates the total prevalence of 6.3%

Weighted Mean Hearing Thresholds by Frequency (N = 3299)

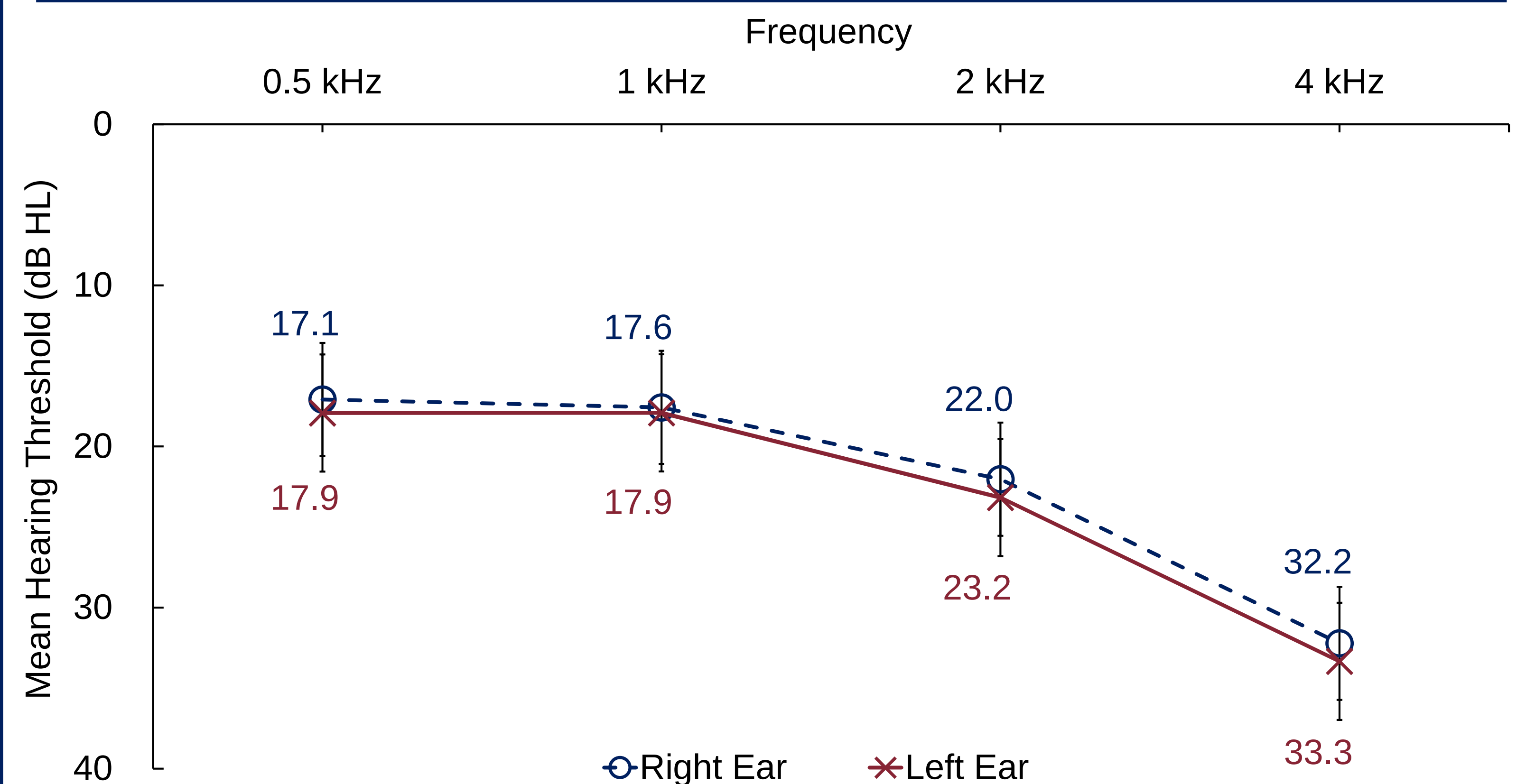


Figure 2. Left-ear thresholds were higher than right-ear thresholds across all frequencies, with the interaural difference more pronounced at 2 and 4 kHz.

Frequency-Specific Asymmetry (n = 207)

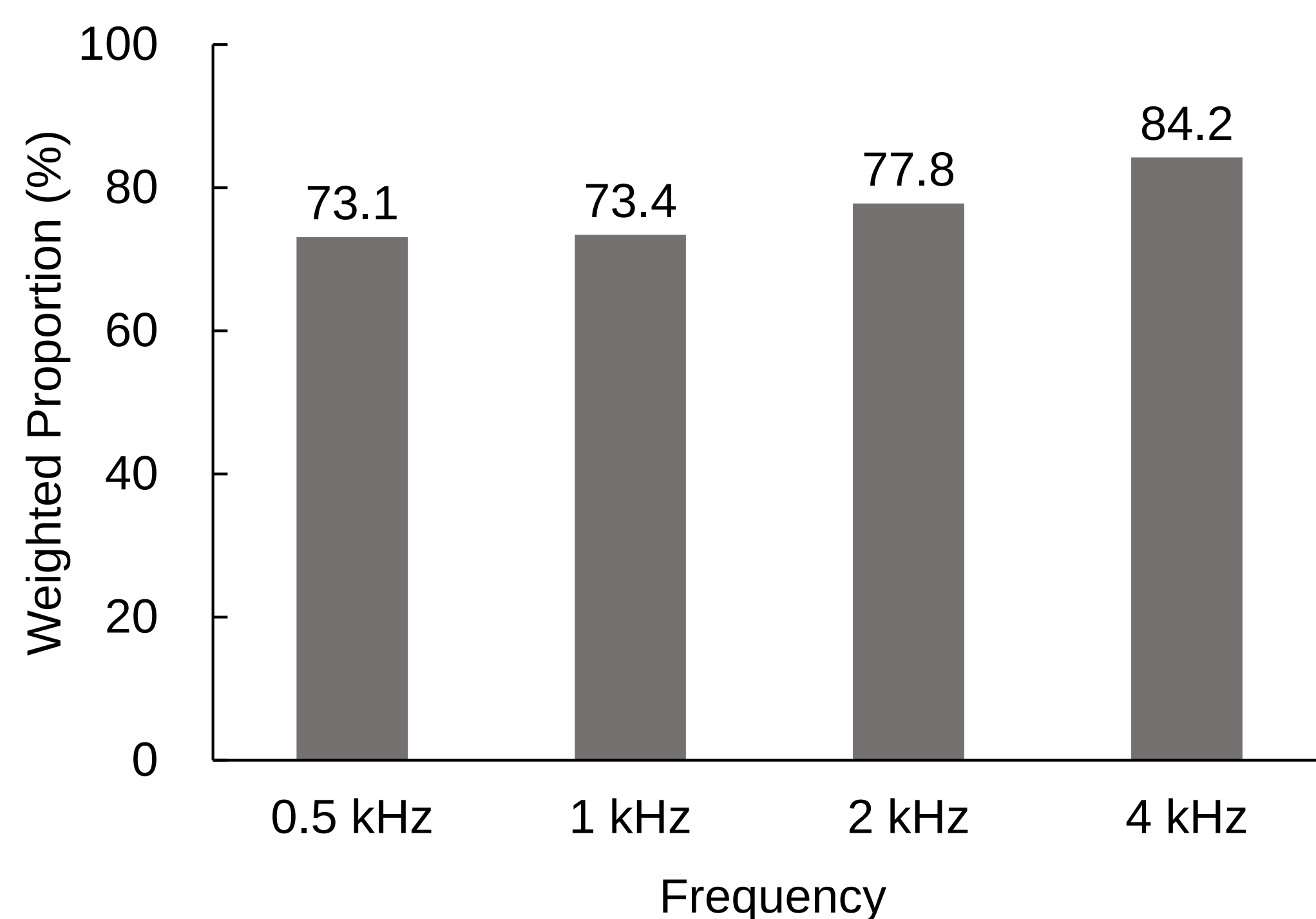


Figure 3. Asymmetry increased with frequency. Proportions are not mutually exclusive and do not sum to 100%, because participants could have asymmetry at more than one frequency.

Laterality of AHL (n = 207)

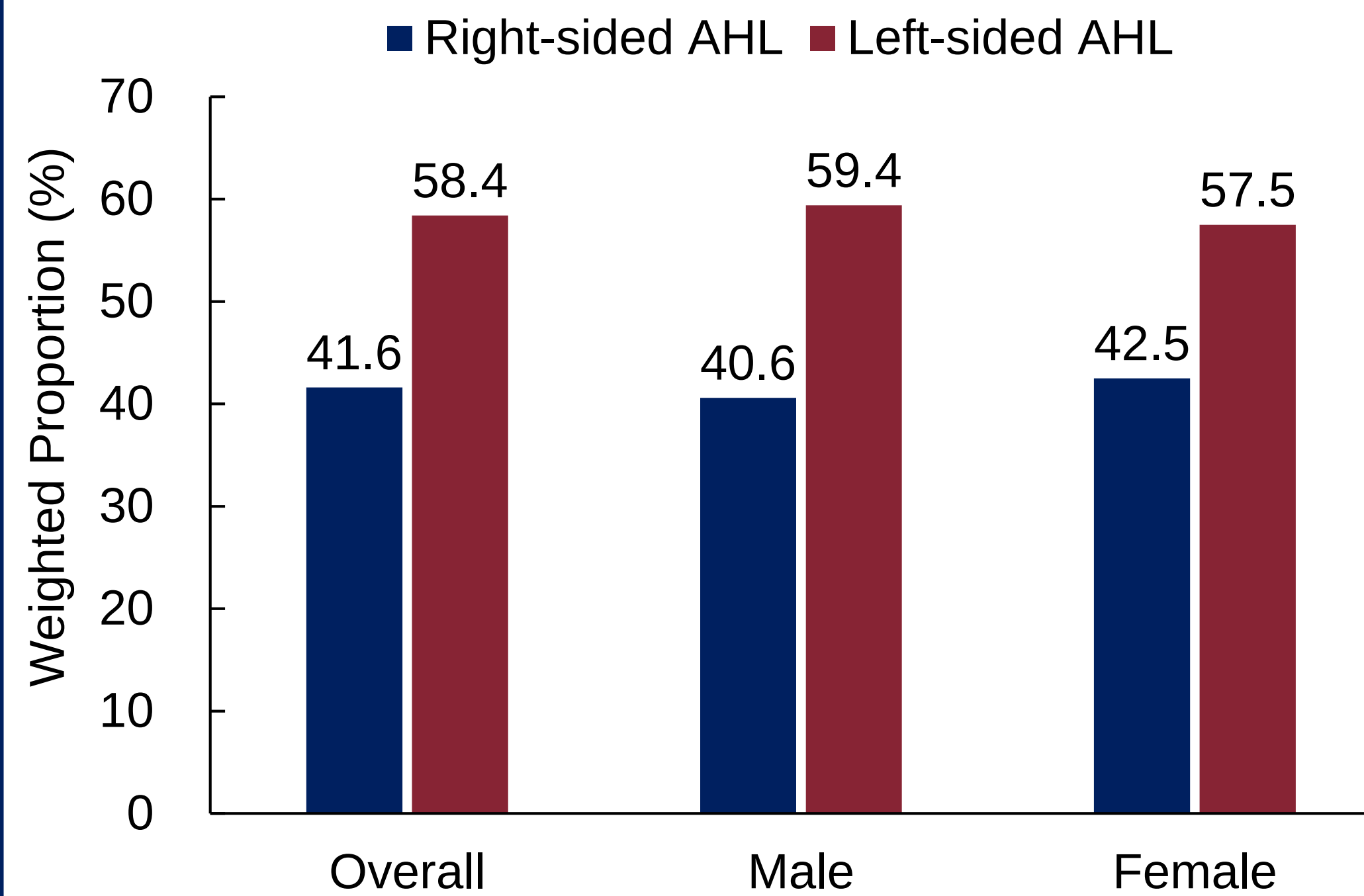


Figure 4. Left-sided AHL was more common than right-sided AHL overall and in both sexes, indicating a consistent left-sided predominance across sex subgroups.

Interaural Differences by Age (n = 207)

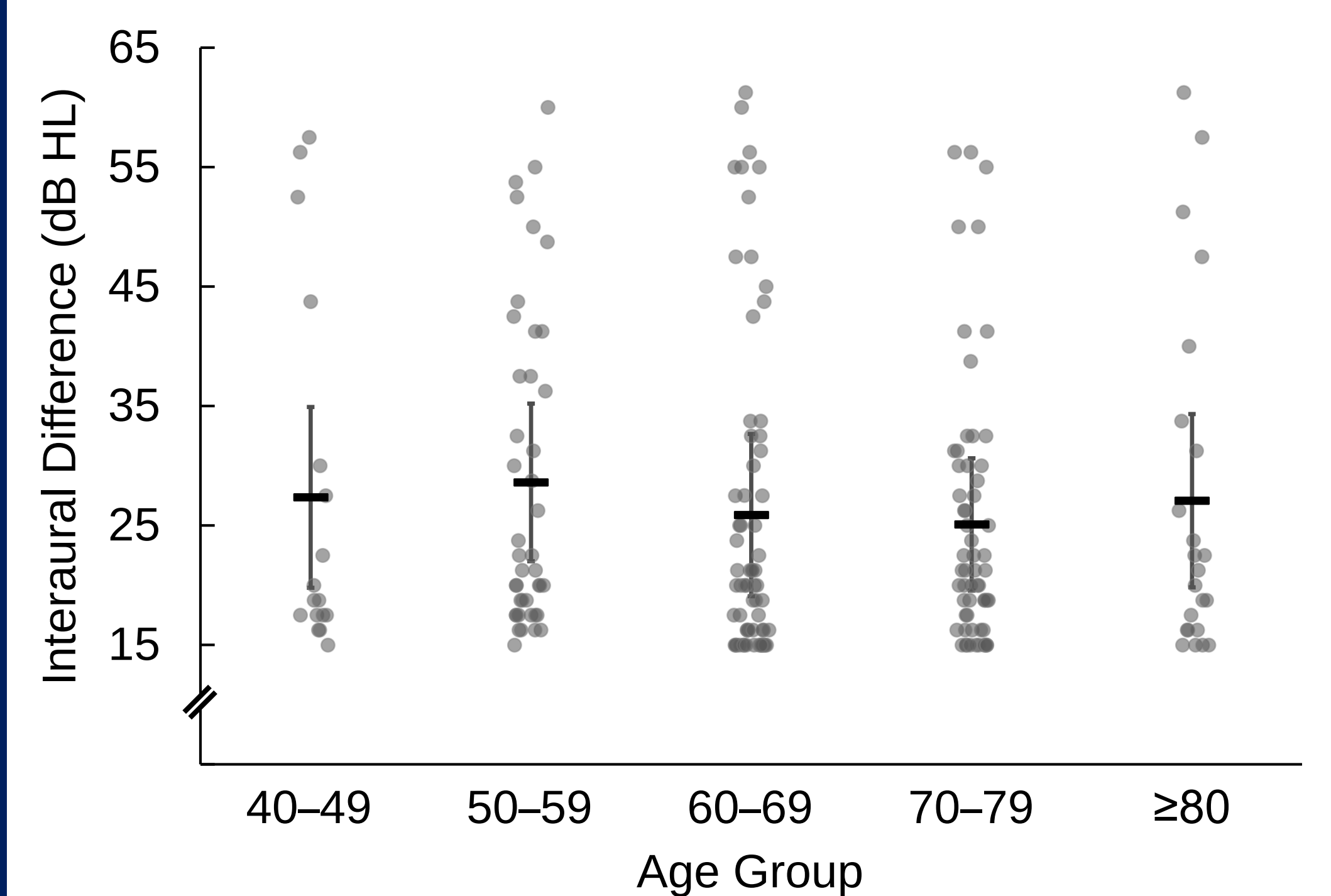


Figure 5. Interaural differences did not differ significantly across age groups, with most cases clustered within the 15–20 dB HL range. Horizontal bars and error bars indicate mean $\pm \frac{1}{2}$ SD.



CONCLUSION

In Korean adults aged ≥ 40 years, AHL affected **6.3%** of the population and generally increased with age overall and in both sexes. AHL was characterized by **higher-frequency asymmetry**, a consistent **left-sided predominance**, and clustering near the lower end of the asymmetry range, with **most cases falling within the 15–20 dB HL range**. These findings suggest that population-level AHL is most often expressed as relatively mild interaural asymmetry rather than marked asymmetry.



RESOURCES



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Access graphs, figures and full-text references

